Cospectral periodic graphs: the spectrum does not determine the graph

Jiaqi Yu*

September 21, 2026

Abstract

Two infinite, locally finite, vertex-transitive graphs can be cospectral without being isomorphic, and we exhibit such a pair inside one natural family. Let $V_K = \{x \in \mathbb{Z}^m : \sum_i x_i \in \{0,1\}\}$ and let $K(m)$ be the graph on V_K with edges $\{x, x \pm e_i\}$; let $\widetilde{S}_m$ be the chamber graph of the affine Weyl group of type $\widetilde{A}_{m-1}$, i.e. a Cayley graph of $Q^\vee \rtimes S_m$. Both are m-regular, infinite, vertex-transitive and of periodic rank $m-1$. We prove $\operatorname{Spec} K(m) = [-m, m]$, and quote from the companion book that $\operatorname{Spec} \widetilde{S}_m = [-m, m]$ for $3 \le m \le 5$. Yet $\operatorname{girth} K(m) = 6$ while $\operatorname{girth} \widetilde{S}_m = 4$ for $m \ge 4$; so for $m = 4$ and $m = 5$ the two graphs are cospectral, of the same degree and the same periodic rank, and not isomorphic. Consequently no spectral invariant, read as a set, can decide the identity of a periodic graph within this family.

Keywords: periodic graph; Cayley graph; spectrum; girth; Floquet decomposition; affine Weyl group.

MSC 2020: 05C25 (primary); 05C50, 05C38, 20F55 (secondary).

1 Introduction

For finite graphs the spectrum is a rich but incomplete invariant: cospectral non-isomorphic graphs are abundant, with small explicit examples. For *infinite, periodic* graphs the situation is less familiar, and there is a natural temptation to hope that the extra structure — a free abelian group acting cocompactly — makes the spectrum a complete invariant. The purpose of this note is to record an explicit counterexample, inside a family of periodic graphs that arises from a single root system.

1.1 The two graphs

Fix $m \ge 3$ and let $e_1, \dots, e_m$ be the standard basis of $\mathbb{Z}^m$. Put

$$V_K := \Big\{ x \in \mathbb{Z}^m : \ \sum_{i=1}^m x_i \in \{0,1\} \Big\}. \tag{1}$$

Definition 1.1 (the folded graph $K(m)$)**.** $K(m)$ is the graph with vertex set V_K in which x and y are adjacent when $y = x \pm e_i$ for some i.

*Tongji University. 2553059@tongji.edu.cn

Definition 1.2 (the chamber graph $\widetilde{S}_m$)**.** Let $Q^\vee = \{x \in \mathbb{Z}^m : \sum_i x_i = 0\} \cong \mathbb{Z}^{m-1}$ be the coroot lattice of A_{m-1} and let $W_a(\widetilde{A}_{m-1}) = Q^\vee \rtimes S_m$ be the affine Weyl group, acting on $\mathbb{Z}^m$ by $x \mapsto w(x) + \lambda$ with $w \in S_m$, $\lambda \in Q^\vee$. Let $s_1, \dots, s_{m-1}$ be the simple reflections of the finite Weyl group and $s_0 = \big(s_\theta,\, \theta\big)$ the affine reflection, where $\theta = e_1 - e_m$ and s_θ is the transposition of coordinates 1 and m. The chamber graph $\widetilde{S}_m := \mathrm{Cay}\big(W_a, \{s_0, s_1, \dots, s_{m-1}\}\big)$ is the graph on the chambers in which two chambers are adjacent when they share a wall.

Both graphs are m-regular, infinite, locally finite and vertex-transitive, and both carry a free action of a rank-$(m-1)$ free abelian group with finite quotient: $Q^\vee := \{x \in \mathbb{Z}^m : \sum_i x_i = 0\}$ acts on V_K by translation for $K(m)$, and it acts on W_a by left translation for $\widetilde{S}_m$. We record this in Section 2 and use it to define the spectrum.

1.2 Main results

Theorem 1.3 (spectrum of $K(m)$)**.** *Let $m \geq 3$. Then* $\operatorname{Spec} K(m) = [-m, m]$*, where the spectrum of a periodic graph is the closure of the union of the spectra of its Floquet (Bloch) matrices over the dual torus, as recalled in Section 2.*

Theorem 1.4 (girths)**.** *Let $m \geq 3$. Then* $\operatorname{girth} K(m) = 6$*. For the chamber graph,* $\operatorname{girth} \widetilde{S}_m = 6$ *when $m = 3$ and* $\operatorname{girth} \widetilde{S}_m = 4$ *for every $m \geq 4$.*

Theorem 1.5 (the pair is cospectral and non-isomorphic)**.** *Let $m \in \{4, 5\}$. Then $K(m)$ and $\widetilde{S}_m$ are both m-regular, infinite, vertex-transitive, of periodic rank $m-1$, and*

$$\operatorname{Spec} K(m) = [-m, m] = \operatorname{Spec} \widetilde{S}_m,$$

but $\operatorname{girth} K(m) = 6 \neq 4 = \operatorname{girth} \widetilde{S}_m$*. In particular $K(m) \not\cong \widetilde{S}_m$: the spectrum does not determine the graph.*

Remark 1.6. The spectrum of $\widetilde{S}_m$ is taken from the companion book [6, Thm. 6.5], where $\operatorname{Spec} \widetilde{S}_m = [-m, m]$ is proved for $3 \leq m \leq 5$ (for $m \geq 6$ the inclusion $\operatorname{Spec} \widetilde{S}_m \subseteq [-m, m]$ and the two endpoints $\pm m$ are proved there, and the equality is open). Our Theorem 1.5 is therefore stated for $m = 4, 5$, matching the known range; the case of $\widetilde{S}_m$ for $m \geq 6$ is an open problem that would extend it. The spectrum of $K(m)$ is proved here for all $m \geq 3$.

Remark 1.7 (why the spectrum alone cannot decide identity). Theorem 1.5 is a negative result about one proposed procedure: any identification that uses only the spectrum *as a set* — in particular $\lambda_{\max}$, $\lambda_{\min}$, or the statement that the spectrum is an interval — assigns the same value to $K(m)$ and $\widetilde{S}_m$ and therefore cannot separate them. We do *not* claim that no spectral invariant can: finer data, such as the spectral measure or the density of states, may well distinguish the pair, and the computations of Section 5 do not settle that question. What separates the pair in this note is the girth, an elementary metric invariant.

Example 1.8 (the smallest case)**.** For $m = 4$: $K(4)$ and $\widetilde{S}_4$ are both 4-regular, infinite, vertex-transitive, $\operatorname{Spec} K(4) = \operatorname{Spec} \widetilde{S}_4 = [-4, 4]$, and $\operatorname{girth} K(4) = 6$ while $\operatorname{girth} \widetilde{S}_4 = 4$.

A 6-cycle in $K(4)$, written in the coordinates $x = (x_1, x_2, x_3, x_4) \in V_K$, is

$$(0,0,0,0),\ (0,0,0,1),\ (-1,0,0,1),\ (-1,0,1,1),\ (-1,0,1,0),\ (0,0,1,0),$$

returning to $(0,0,0,0)$. The successive differences $e_4,\ -e_1,\ e_3,\ -e_4,\ e_1,\ -e_3$ are signed standard basis vectors, so consecutive vertices are adjacent; the six vertices are pairwise distinct; and each has coordinate sum in $\{0, 1\}$ $(0, 1, 0, 1, 0, 1)$. Hence they form a 6-cycle.

A 4-cycle in $\widetilde{S}_4$ is the square 1, s_1, s_1s_3, s_3: the nodes 1 and 3 are not adjacent in the Coxeter cycle of type $\widetilde{A}_3$, so s_1 and s_3 commute and the walk closes, $1 \to s_1 \to s_1s_3 \to s_3 \to 1$; its four vertices are distinct in W_a, so it is a 4-cycle. (The pair s_1, s_2 would not do: adjacent nodes do not commute.)

1.3 Method, and what is not claimed

The paper has two halves. The first (Section 3) constructs $K(m)$'s periodic structure from Definition 1.1 and proves $\operatorname{Spec} K(m) = [-m, m]$ by reducing it to a single statement about sums of m unit complex numbers; the periodicity is used only through the Floquet decomposition of Section 2, whose short proof we include. The second half (Section 4) determines the girth of the chamber graph, which is an exercise in Coxeter combinatorics.

Novelty is not claimed for the phenomenon that cospectral non-isomorphic graphs exist, nor for the Floquet decomposition. What is exhibited here is a pair of non-isomorphic periodic graphs that are cospectral *and* share every crude invariant one would try first — degree, vertex-transitivity, periodic rank, and the two spectral endpoints — and that are separated only by the girth.

1.4 Organisation

Section 2 fixes the notion of a periodic graph and its spectrum and proves the Floquet decomposition. Section 4 treats the chamber graph and proves Theorem 1.4, and Section 5 records the verification. Section 6 discusses the open case $m \geq 6$.

2 Periodic graphs and their spectra

Definition 2.1 (periodic graph)**.** Let X be a locally finite, connected, simple graph and let $\Gamma \leq \operatorname{Aut}(X)$ be a free abelian group acting freely on $V(X)$ with finitely many orbits. We call X a Γ*-periodic graph* and call $\operatorname{rank}\Gamma$ its *periodic rank*.

We shall use that both graphs of Section 1.1 are periodic in this sense.

Lemma 2.2. *$K(m)$ is $Q^\vee$-periodic of rank $m-1$ with exactly two orbits on V_K, namely $\{\sum_i x_i = 0\}$ and $\{\sum_i x_i = 1\}$; and $\widetilde{S}_m$ is $Q^\vee$-periodic of rank $m-1$ with $|W_a/Q^\vee| = m!$. Both graphs are m-regular and vertex-transitive.*

Proof. $Q^\vee = \{x \in \mathbb{Z}^m : \sum_i x_i = 0\}$ preserves the value of $\sum_i x_i$, hence preserves V_K and each of its two levels; it acts freely on $\mathbb{Z}^m$. Since $\sum_i x_i \in \{0,1\}$ on V_K and $Q^\vee$ is the full stabiliser of the two levels, the orbits are exactly the two level sets, so $K(m)$ is $Q^\vee$-periodic with two orbits. Regularity: the neighbours of $x \in V_K$ are the $x \pm e_i$ that lie in V_K; since $\sum_i (x \pm e_i) = \sum_i x \pm 1$, for each i exactly one of $x + e_i, x - e_i$ has its sum in $\{0,1\}$, so every vertex has exactly m neighbours and $K(m)$ is m-regular. Finally $x \mapsto x + \lambda$ ($\lambda \in Q^\vee$) is a graph automorphism, so $K(m)$ is vertex-transitive. The statements for $\widetilde{S}_m$ are the standard semidirect-product structure of the affine Weyl group, $W_a = Q^\vee \rtimes S_m$ with $Q^\vee$ normal and $W_a/Q^\vee \cong S_m$ [1, Ch. IV–VI], together with the fact that a Cayley graph is regular of degree $|S|$ and vertex-transitive. □

Remark 2.3. The choice of translation lattice matters. A lattice can serve as a period lattice only if it preserves V_K, and that already forces it to lie in $Q^\vee$: if v and $e_1 + v$ both lie in V_K then $\sum_i v_i \in \{0,1\}$ and $1 + \sum_i v_i \in \{0,1\}$, whence $\sum_i v_i = 0$. So $Q^\vee$ is the *maximal* period lattice of $K(m)$ and every other legitimate choice is a sublattice $\Lambda \subseteq Q^\vee$ of finite index. For example

$\Lambda := mQ^\vee = \{mv : \sum_i v_i = 0\}$ has index m^{m-1} in $Q^\vee$ and gives a cell with $2m^{m-1}$ vertices (the classes are indexed by $(x_1, \dots, x_{m-1})$ mod m and by $\sum_i x_i \in \{0, 1\}$), and a Floquet matrix of that size. That is legitimate but harder; we therefore use the maximal lattice $Q^\vee$. Its cell is the two-vertex graph of Lemma 3.1.

Definition 2.4 (spectrum of a periodic graph)**.** Let X be Γ-periodic with finite quotient X/Γ (a multigraph, the *cell*), and let A be the adjacency operator of X on $\ell^2(V(X))$. For a character $\chi \in \widehat{\Gamma} = \mathbb{T}^d$, $d = \operatorname{rank}\Gamma$, let A_χ be the *Floquet matrix*: the matrix indexed by $V(X/\Gamma)$ in which the entry from the class of x to the class of y is $\sum \chi(\lambda)$, the sum over all λ and all adjacencies $x \sim y + \lambda$ that induce the given class-to-class transition. The *spectrum* of X is

$$\operatorname{Spec} X := \overline{\bigcup_{\chi \in \mathbb{T}^d} \operatorname{Spec} A_\chi}. \tag{2}$$

Proposition 2.5 (Floquet decomposition [3, 2])**.** *With the notation of Definition 2.4,* $\operatorname{Spec} X$ *as defined in* (2) *coincides with the spectrum of the adjacency operator* A *on* $\ell^2(V(X))$*.*

Proof. Γ acts unitarily on $\ell^2(V(X))$ by $\big(U_\lambda f\big)(x) = f(x - \lambda)$, and Γ is abelian, so its spectral decomposition is the direct integral over $\widehat{\Gamma}$ of the fibres $\ell^2(V(X/\Gamma))$; this is the standard Floquet (or Bloch) decomposition, i.e. the direct integral decomposition of the operator, for a free abelian cocompact action [2, §4]. For discrete graphs it is stated in [3, eq. (1.11) and §7] as $\sigma(H) = \bigcup_{\vartheta \in \mathbb{T}^d} \sigma\big(H(\vartheta)\big)$ for the d-periodic Schrödinger operator $H = \Delta + Q$. Concretely, the Fourier transform $\widehat{f}(\chi, y) = \sum_{\lambda \in \Gamma} \chi(\lambda) f(y + \lambda)$, defined first for compactly supported f and extended continuously, intertwines A with $\int^{\oplus} A_\chi \, d\chi$: the entry $A_\chi(y, y')$ is exactly the χ-weighted sum of the translation vectors realising the transition $y \to y'$, by Definition 2.4. Hence A is unitarily equivalent to the direct integral, and the spectrum of a direct integral of finite matrices over a compact space is the closure of the union of the fibre spectra. This is (2). □

Remark 2.6. Definition 2.4 is phrased for the cell as a multigraph, because a cell can have parallel edges (two vertices of the quotient may be joined in more than one way); the Floquet matrices then sum the characters of the several translations realising the same class-to-class transition. For $K(m)$ the cell has two vertices and m parallel edges (Lemma 3.1(iii)), so the multigraph formulation is exactly what makes Definition 2.4 apply, and it covers $\widetilde{S}_m$ as well. We never treat $K(m)$ itself as a multigraph — it is a simple graph by Definition 1.1.

3 The graph $K(m)$ and its spectrum

We keep the notation of Section 2: $Q^\vee = \{x : \sum_i x_i = 0\}$ is the translation lattice, identified with $\mathbb{Z}^{m-1}$ by $x \mapsto (x_1, \dots, x_{m-1})$, and $\chi \in \mathbb{T}^{m-1}$ is a character of $Q^\vee$.

Lemma 3.1 (the cell of $K(m)$)**.** *Let* $Q^\vee = \{x \in \mathbb{Z}^m : \sum_i x_i = 0\}$*, and for* $x \in V_K$ *write*

$$A_a := (a_1, \dots, a_{m-1}, -\textstyle\sum_{i<m} a_i) \in V_K \quad (\textstyle\sum_i x_i = 0), \tag{3}$$

$$B_b := (b_1, \dots, b_{m-1}, 1 - \textstyle\sum_{i<m} b_i) \in V_K \quad (\textstyle\sum_i x_i = 1), \tag{4}$$

for $a, b \in \mathbb{Z}^{m-1}$*. Then:*

(i) $\{A_a : a \in \mathbb{Z}^{m-1}\}$ *and* $\{B_b : b \in \mathbb{Z}^{m-1}\}$ *are exactly the two* $Q^\vee$*-orbits in* V_K*, and each of them is in bijection with* $\mathbb{Z}^{m-1}$ *via* $x \mapsto (x_1, \dots, x_{m-1})$*;*

(ii) the edges of $K(m)$ are

$$A_a \sim B_{a+e_i}\ (1 \le i \le m-1), \qquad A_a \sim B_a, \qquad B_b \sim A_{b-e_i}\ (1 \le i \le m-1), \qquad B_b \sim A_b; \tag{5}$$

(iii) consequently the cell $V_K/Q^\vee$ of Definition 2.4 is a two-vertex graph with m parallel edges, of voltages $0, e_1, \dots, e_{m-1} \in Q^\vee$, and the Floquet matrix is

$$A_\chi = \begin{pmatrix} 0 & S(\chi) \\ \overline{S(\chi)} & 0 \end{pmatrix}, \qquad S(\chi) := 1 + z_1 + \cdots + z_{m-1}, \quad z_i := \chi(e_i). \tag{6}$$

Hence $\operatorname{Spec} A_\chi = \{\pm|S(\chi)|\}$.

Proof. (i) $Q^\vee$ preserves V_K because it preserves $\sum_i x_i$. The vector A_a has sum 0 and the vector B_b has sum 1, so A_a and A_b (resp. B_a and B_b) are in the same orbit if and only if their first $m-1$ coordinates agree; and the first $m-1$ coordinates are already an injective invariant, since the last coordinate of a vector of V_K is determined by the sum, which is 0 or 1. Hence the two orbits, each in bijection with $\mathbb{Z}^{m-1}$.

(ii) Let $x = A_a$, and add $\pm e_i$. Adding $+e_i$ for $i \le m-1$ gives a vector whose sum is 1, i.e. a B-vector, with first $m-1$ coordinates $a + e_i$; adding $+e_m$ gives sum 1 and first $m-1$ coordinates a; adding $-e_i$ or $-e_m$ gives sum -1 or 0, and a vector of sum -1 is not in V_K, while a vector of sum 0 with the same first $m-1$ coordinates is A_a itself. So the neighbours of A_a in V_K are exactly the B's displayed in (5). The computation from a B-vector is the same with the signs reversed.

(iii) By (i) a cell is described by two vertices A and B, and by (ii) the edges joining them correspond to the translations $0, e_1, \dots, e_{m-1}$ taking A_0 to $B_0 + e_i$ or B_0. Hence in the cell there are m parallel edges A–B with these voltages. For a character χ of $Q^\vee \cong \mathbb{Z}^{m-1}$, the Floquet matrix of Definition 2.4 has off-diagonal entries the sum of the χ-weights of these m edges, namely $S(\chi) = \chi(0) + \sum_{i<m} \chi(e_i) = 1 + \sum_{i<m} z_i$, and the conjugate in the opposite direction; so A_χ is as displayed. The eigenvalues of $\begin{pmatrix} 0 & S \\ \overline{S} & 0 \end{pmatrix}$ are $\pm|S|$: its square is $|S|^2 I$. □

Remark 3.2 (bipartition). Lemma 3.1(ii) shows that every edge of $K(m)$ joins an A-vertex to a B-vertex; hence $K(m)$ is bipartite with the two classes $\{\sum_i x_i = 0\}$, $\{\sum_i x_i = 1\}$. It is m-regular: each A_a has the m neighbours $B_{a+e_1}, \dots, B_{a+e_{m-1}}, B_a$, and each B_b has m neighbours as well.

Proof of Theorem 1.3. By Lemma 3.1(iii) and Definition 2.4,

$$\operatorname{Spec} K(m) \;=\; \overline{\bigcup_{\chi \in \mathbb{T}^{m-1}} \operatorname{Spec} A_\chi} \;=\; \overline{\bigcup_{\chi \in \mathbb{T}^{m-1}} \{\pm|S(\chi)|\}} \;=\; \overline{\pm|S(\mathbb{T}^{m-1})|}, \tag{7}$$

where $S(\chi) = 1 + z_1 + \cdots + z_{m-1}$ with $z_i = \chi(e_i)$, and $|S(\mathbb{T}^{m-1})| := \{|S(\chi)| : \chi \in \mathbb{T}^{m-1}\}$.

(a) $|S(\mathbb{T}^{m-1})| = [0, m]$. $S(\chi)$ is the sum of the m unit complex numbers $1, z_1, \dots, z_{m-1}$, so $|S(\chi)| \le m$ for every χ by the triangle inequality, with equality at $z_1 = \cdots = z_{m-1} = 1$. Also $|S| = 0$ is attained: for $m = 2$ take $z_1 = -1$; for $m = 3$ take $z_1 = e^{2\pi i/3}$, $z_2 = e^{4\pi i/3}$, the two primitive cube roots, whose sum is -1; and for $m \ge 4$ take $z_1 = -1$ together with *all* $m-2$ of the $(m-2)$-th roots of unity $z_2, \dots, z_{m-1}$, whose sum is 0. In every case $S(\chi) = 1 + \sum_i z_i = 0$. Finally $\chi \mapsto |S(\chi)|$ is continuous and $\mathbb{T}^{m-1}$ is connected, so $|S(\mathbb{T}^{m-1})|$ is a connected subset of $[0, m]$ containing 0 and m, hence equals $[0, m]$.

(b) Conclusion. Substituting into (7), $\operatorname{Spec} K(m) = \overline{[-m, m]} = [-m, m]$. □

Remark 3.3. Two points in the proof deserve a word. The set $|S(\mathbb{T}^{m-1})|$ is connected because it is the continuous image of the connected torus — this is what rules out a spectral gap, and it is the only place where the topology of the torus is used. And the witness for $|S| = 0$ is explicit at every m, so the endpoint 0 of the band need not be searched for numerically (which matters because a coarse grid of characters misses it: on a 60-point grid in each angle the sampled minimum of $|S|$ for $m = 4$ is about 0.04 rather than 0).

Remark 3.4. The proof also yields the bipartition of $K(m)$: the two coroot levels are independent sets, since every edge changes $\sum_i x_i$ by ± 1; explicitly, the bipartition is $\{A_a\} \cup \{B_b\}$ in the notation (3).

Remark 3.5 (the edge count). Taking each of the $2m$ directed steps $\pm e_i$ of $K(m)$ to contribute a term $\chi(\pm e_i)$ to $S(\chi)$ gives $S(1) = 2(m-1) + 1 = 2m - 1$ and $\lambda_{\max} = 2m - 1$. That is also wrong: by (5) the two signs lead to the *same* edge of the cell out of a vertex of the cell, so the cell has one edge per direction and $S(1) = m$. For $m = 4$ the false answer is 7, against the true degree 4. The mistake is caught at once by comparing $\lambda_{\max}$ with the degree. The other way to get the cell wrong — the choice of the period lattice — is recorded in Remark 2.3.

4 The chamber graph $\widetilde{S}_m$ and the girths

Proof of Theorem 1.4, $K(m)$. Let $x \in V_K$ and let $x \sim x + se_i \sim y$ be two steps. If the second step undoes the first, the walk returns to x. Otherwise the second step is $x + se_i + te_j$ with $(t, j) \neq (-s, i)$, and a triangle occurs precisely when $x + se_i + te_j = x$, i.e. $se_i + te_j = 0$, which forces $j = i$ and $t = -s$, the excluded case. So $K(m)$ has no triangle. A 4-cycle would give four steps $\sum_k s_k e_{i_k} = 0$ with no two consecutive steps inverse and the four intermediate vertices distinct; but $\sum_k s_k e_{i_k} = 0$ forces the multiset $\{(s_k, i_k)\}$ to cancel in pairs, and with four steps the only possibility is $s_1 e_{i_1} + s_2 e_{i_2} = -s_3 e_{i_3} - s_4 e_{i_4}$, i.e. $\{i_1, i_2\} = \{i_3, i_4\}$, which makes the walk backtrack. Hence girth $K(m) \geq 6$, and the pattern x, $x + e_1$, $x + e_1 - e_2$, $x + e_1 - e_2 + e_3$, $x - e_2 + e_3$, $x + e_3$, x — the 6-cycle of Example 1.8, which uses only the three directions e_1, e_2, e_3 and so applies for every $m \geq 3$ — gives girth $K(m) = 6$. □

Proof of Theorem 1.4, $\widetilde{S}_m$. $\widetilde{S}_m$ is the Cayley graph of W_a with respect to $S = \{s_0, \ldots, s_{m-1}\}$, and S consists of involutions, so there are no loops; since $s_i \neq s_j$ for $i \neq j$ and $s_i^{-1} = s_i$, there are no double edges either. We exhibit a 4-cycle for $m \geq 4$ from a pair of commuting simple reflections, and then treat $m = 3$, where no 4-cycle exists.

No triangles. A triangle would be three distinct generators with $s_i s_j s_k = 1$, i.e. $s_i s_j = s_k$. But in a Coxeter group the only relations of rank 2 among distinct simple reflections are $(s_i s_j)^{m_{ij}} = 1$ with $m_{ij} \geq 2$, so $s_i s_j$ has order $m_{ij} \geq 2$ and cannot be a simple reflection when $m_{ij} \geq 3$ (its own order would be 2); and if $m_{ij} = 2$ then $s_i s_j = s_j s_i \neq s_k$ for $k \notin \{i, j\}$. Hence no triangle: girth $\widetilde{S}_m \geq 4$.

4-cycles are exactly the commuting pairs. Let $s_{i_1} s_{i_2} s_{i_3} s_{i_4} = 1$ be a 4-cycle through 1, so its four partial products are distinct; in particular consecutive letters differ. Multiplying by $s_{i_3} s_{i_4}$ gives $s_{i_1} s_{i_2} = s_{i_4} s_{i_3}$, an equality between two products of two distinct simple reflections, and it forces $\{i_1, i_2\} = \{i_4, i_3\}$: if the two pairs share exactly one index, multiplying the equality by that simple reflection turns it into $s_a s_b s_a = s_c$ with $a \neq b$, whose left side is either s_b — when $m_{ab} = 2$, and then the two pairs coincide after all — or a reduced word of length 3 when $m_{ab} \geq 3$; and if the two pairs are disjoint, then the relation gives $s_{i_2} s_{i_3} s_{i_4} = s_{i_1}$, a product of three distinct simple reflections on the left against a simple reflection on the right. So either $i_4 = i_1$, $i_3 = i_2$ (then the

word is $s_i s_j s_j s_i$, whose second and third vertices coincide, so it is not a cycle) or $i_4 = i_2$, $i_3 = i_1$, i.e. $(s_i s_j)^2 = 1$, which holds exactly when $m_{ij} = 2$; and then the walk

$$1 \xrightarrow{s_i} s_i \xrightarrow{s_j} s_i s_j \xrightarrow{s_i} s_j \xrightarrow{s_j} 1$$

is a 4-cycle, its vertices being pairwise distinct ($s_i, s_j \neq 1$, $s_i s_j \neq s_i, s_j$ and $s_i s_j \neq 1$). In type $\widetilde{A}_{m-1}$ two distinct simple reflections commute exactly when their nodes are non-adjacent, and such a pair exists exactly when $m \geq 4$. Hence girth $\widetilde{S}_m = 4$ for every $m \geq 4$, and there is no 4-cycle for $m = 3$.

The case $m = 3$. Here $\widetilde{A}_2$ has all three pairs of nodes adjacent, so no two simple reflections commute and there is no 4-cycle; the graph has no triangle either, and the sign character of W_a (which sends every s_i to -1) makes every closed walk of even length, so girth $\widetilde{S}_3 \geq 6$. The braid cycle $1,\ s_i,\ s_i s_j,\ s_i s_j s_i,\ (s_i s_j)^2,\ s_j,\ 1$ — the six elements of the rank-two parabolic subgroup $\langle s_i, s_j \rangle \cong S_3$ — is a 6-cycle, so girth $\widetilde{S}_3 = 6$. This is confirmed by the computation of Section 5. □

Proof of Theorem 1.5. Combine Lemma 2.2, Theorem 1.3, the quoted $\operatorname{Spec} \widetilde{S}_m = [-m, m]$ [6, Thm. 6.5], and Theorem 1.4. Since isomorphic graphs have equal girth, $K(m) \ncong \widetilde{S}_m$ for $m = 4, 5$. □

5 Numerical verification

The two scripts `code/verify_all.py` and `code/spec_K_proof.py`, distributed with the source, together check the numerically falsifiable statements listed below; their acceptance tests are the definitions, and both end with `ALL CHECKS PASS`.

- **Size and regularity (Lemma 2.2).** For $m = 3, 4, 5$ the class count is $|V_K/\Lambda| = 2m^{m-1}$ (18, 128, 1250) for the sublattice $\Lambda = mQ^\vee$ of index m^{m-1} in $Q^\vee$, and every vertex has exactly m neighbours.
- **Spectrum (Theorem 1.3).** Two independent checks. (i) The identity character gives the adjacency matrix, whose extremes are exactly $\pm m$ for $m = 3, 4, 5$ (to machine precision). (ii) The Floquet matrices are diagonalised on a grid of characters; the sampled eigenvalues stay inside $[-m, m]$, fill it, and a continuous eigenvalue path from χ_0 to a generic character sweeps the whole interval. Neither check is used as a proof: the proof of Theorem 1.3 is the argument of Section 3.
- **Girth (Theorem 1.4).** $K(m)$: BFS in a box gives girth 6 for $m = 3, 4, 5$. $\widetilde{S}_m$: the affine Weyl group is realised as $\{(w, \lambda)\}$ with the group law $(w, \lambda)(v, \mu) = (wv, \lambda + w(\mu))$, the ball of radius 6 is built from the Coxeter presentation, and the girth computed inside it is 6 for $m = 3$ and 4 for $m = 4, 5$.
- **Non-isomorphism (Theorem 1.5).** Same degree and different girth for $m = 4$ (6 versus 4) and $m = 5$ (6 versus 4).

No numerical output is used as a proof: every statement is proved in the text.

6 Discussion and open problems

The gap at $m \geq 6$. The equality $\operatorname{Spec} \widetilde{S}_m = [-m, m]$ is known for $3 \leq m \leq 5$ [6, Thm. 6.5]; for $m \geq 6$ only the inclusion into $[-m, m]$ and the two endpoints are proved there. Closing this gap for $m \geq 6$

would extend Theorem 1.5 to all $m \geq 4$. The obstruction is that the cell of $\widetilde{S}_m$ has $m!$ vertices, that of $K(m)$ two: for $K(m)$ the Floquet matrix is 2×2 and the equality $\operatorname{Spec} K(m) = [-m, m]$ reduces to the single statement that $|S(\chi)|$ attains 0, which has an explicit witness, whereas for $\widetilde{S}_m$ one must show that two branches meet, with no explicit witness available.

What other invariants separate the pair? The degree, the periodic rank, the two spectral endpoints and bipartiteness all agree for the pair, so none of them can identify it; whether finer spectral data, such as the density of states, also agree is left open in Remark 1.7. Girth separates them. It would be interesting to know the smallest metric invariant that does, and whether the pair can be separated by a finite computation on the cell (the answer is yes if and only if the two cells are not isomorphic as voltage graphs).

Cospectrality in wider families. The same mechanism — a Floquet matrix whose spectrum is $\pm|S(\chi)|$ with $|S|$ sweeping an interval — produces cospectral graphs whenever two periodic graphs of the same degree have the same interval but different local structure. The pair here is the smallest instance we know in this root-system family.

Prior art. Isospectrality is classical for finite graphs, and for *operators* the phenomenon is known in the periodic setting: Floquet-isospectral Schrödinger operators on periodic graphs exist [4], with the strongest recent statement being the construction of a periodic potential Floquet-isospectral to the zero potential [5]. Those results concern operators $\Delta + Q$ on a *fixed* graph and show that the potential is not determined by the Floquet spectrum. The statement here is of a different kind: the two graphs are different, and they are distinguished by the girth, so the failure of spectral determination is a failure of the graph's identity and not of an auxiliary potential. We have not found a printed pair of vertex-transitive periodic graphs that is cospectral for the reasons exhibited here.

Provenance of $\widetilde{S}_m$'s spectrum. It is stated as a quoted theorem rather than reproved here, and this is flagged explicitly in Remark 1.6; an independent proof would make Theorem 1.5 self-contained.

References

[1] N. Bourbaki, *Lie Groups and Lie Algebras, Chapters IV–VI*, Elements of Mathematics, Springer, Berlin, 2002.

[2] P. Kuchment, *An overview of periodic elliptic operators*, Bull. Amer. Math. Soc. (N.S.) **53** (2016), no. 3, 343–414. (§4, and §4.3 in particular, is the Floquet transform and the direct integral decomposition used in Proposition 2.5.)

[3] E. Korotyaev, N. Saburova, *Schrödinger operators on periodic discrete graphs*, J. Math. Anal. Appl. **420** (2014), 576–611; arXiv:1307.1841. (The Floquet decomposition $\sigma(H) = \bigcup_{\vartheta \in \mathbb{T}^d} \sigma(H(\vartheta))$ used in Proposition 2.5 is stated there in §7.)

[4] W. Liu, *Floquet isospectrality for periodic graph operators*, J. Differential Equations **374** (2023), 642–657.

[5] M. Faust, J. Liu, et al., *Floquet isospectrality of the zero potential for discrete periodic Schrödinger operators*, arXiv:2401.09731 (2024).

[6] *Companion book on quotient lattices, tori and periodic graphs*, vol. 11, Theorem 6.5 (spectrum of the chamber graph). Cited for the quoted spectral result; not otherwise used.